Yijun Liang

yliang17@umd.edu | +1-240-481-5811 | Personal Website | Google Scholar | LinkedIn

Ph.D. Student, Department of Computer Science, University of Maryland, College Park

Research Interests: Computer Vision, Visual Perception & Reasoning, Data-Centric AI (Data Synthesis, Benchmark Design)

Education

- **University of Maryland, College Park** — *Aug 2023 – Present*
  Ph.D. in Computer Science. — Maryland, U.S.
- **University of California, Berkeley** — *Aug 2019 – Dec 2020*
  M.Eng. in Industrial Engineering & Operations Research — Berkeley, U.S.
- **Tsinghua University** — *Aug 2015 – Jul 2019*
  B.E. in Industrial Engineering (Operations Research) — Beijing, China

Publications

VideoResearcher: Self-Improving Tool Design for Long-Video Understanding [arXiv] — **arXiv 2026**
Dingqiang Ye, Dongdi Zhao, Kaishen Wang, Qingqiao Hu, Jingchen Sun, **Yijun Liang**, Yuqi Jia, Yiqiao Huang, Yunjie Tian, Jiaxing Zhang, Chuanyang Jin, Ke Zhang, Vishal M. Patel, Di Fu

Where to Look Matters: On-Policy Self-Distillation for Long-Video Understanding [arXiv] — **arXiv 2026**
Kaishen Wang, Dongdi Zhao, **Yijun Liang**, Dingqiang Ye, Ruibo Chen, Heng Huang, Di Fu

Self-Supervised Visual On-Policy Distillation [arXiv] — **arXiv 2026**
Yijiang Li, **Yijun Liang**, Yunjie Tian, Bingyang Wang, Ke Zhang, Zhenfei Yin, Di Fu, Philip Torr, Nuno Vasconcelos

On-Policy Self-Distillation without Any Supervision [arXiv] — **arXiv 2026**
Yijiang Li, Bingyang Wang, **Yijun Liang**, Yunjie Tian, Di Fu, Nuno Vasconcelos

Visual Contrastive Self-Distillation [arXiv] — **arXiv 2026**
Yijun Liang, Yunjie Tian, Yijiang Li, Yuqi Jia, Furong Huang, Tianyi Zhou, Di Fu

Self-Evolving Visual Questioner [arXiv] — **arXiv 2026**
Yijun Liang, Hengguang Zhou, Ming Li, Lichen Li, Cho-Jui Hsieh, Tianyi Zhou

LLMs Struggle to Measure What Distinguishes Students of Different Proficiency Levels: A Study of Item Discrimination in Reading Comprehension Assessment [arXiv] — **arXiv 2026**
Han Chen, Ming Li, Chenguang Wang, **Yijun Liang**, Dawei Zhou, Hong Jiao, Tianyi Zhou

Do Prompt-Elicited Trajectories Reflect Training-Time Reward Hacking? A Systematic Study on Monitoring Training-Time Reward Hacking in Code Generation [arXiv] — **arXiv 2026**
Lichen Li, Hengguang Zhou, **Yijun Liang**, Tianyi Zhou, Cho-Jui Hsieh

V-REX: Benchmarking Exploratory Visual Reasoning via Chain-of-Questions [arXiv] — **ECCV 2026**
Chenrui Fan*, **Yijun Liang***, Shweta Bhardwaj*, Kwesi Cobbina, Ming Li, Tianyi Zhou

History-Conditioned Spatio-Temporal Visual Token Pruning for Efficient Vision-Language Navigation [arXiv] — **IROS 2026**
Qitong Wang, **Yijun Liang**, Ming Li, Tianyi Zhou, Christopher Rasmussen

ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness [arXiv] — **NeurIPS 2025**
Yijun Liang*, Ming Li*, Chenrui Fan, Ziyue Li, Dang Nguyen, Kwesi Cobbina, Shweta Bhardwaj, Jiuhai Chen, Fuxiao Liu, Tianyi Zhou

Diffusion Curriculum: Synthetic-to-Real Generative Curriculum Learning via Image-Guided Diffusion [arXiv] — **ICCV 2025**
Yijun Liang*, Shweta Bhardwaj*, Tianyi Zhou

Mosaic-IT: Free Compositional Data Augmentation Improves Instruction Tuning [pdf] — **ACL 2025**
Ming Li, Pei Chen, Chenguang Wang, Hongyu Zhao, **Yijun Liang**, Yupeng Hou, Fuxiao Liu, Tianyi Zhou

CaughtCheating: Is Your MLLM a Good Cheating Detective? Exploring the Boundary of Visual Perception and Reasoning [arXiv] — **arXiv 2025**
Ming Li, Chenguang Wang, **Yijun Liang**, Xiyao Wang, Yuhang Zhou, Xiyang Wu, Yuqing Zhang, Ruiyi Zhang, Tianyi Zhou

PEDANTS: Cheap but Effective and Interpretable Answer Equivalence [arXiv] — **EMNLP 2024**
Zongxia Li, Ishani Mondal, **Yijun Liang**, Huy Nghiem, Jordan Lee Boyd-Graber

GeoDRL: A Self-Learning Framework for Geometry Problem Solving Using Reinforcement Learning in Deductive Reasoning [pdf] **ACL 2023**
Shuai Peng, Di Fu, **Yijun Liang**, Liangcai Gao, Zhi Tang

Compute Like Humans: Interpretable Step-by-Step Symbolic Computation with Deep Neural Network [pdf] **KDD 2022**
Shuai Peng, Di Fu, Yong Cao, **Yijun Liang**, Gu Xu, Liangcai Gao, Zhi Tang

Work Experience

- **Research Scientist Intern** *May 2026 – Aug 2026*
  TikTok, Seattle, WA
  - **Image & Video VLM post-training**: Developed on-policy distillation (OPD) and on-policy self-distillation (OPSD) recipes for image and video vision–language models.
  - **Privileged-information distillation**: Studied training-time privileged views available only to the teacher, replacing uniform KL with per-token weighting so the student is supervised on signal it can ground.
- **Research Scientist Intern** *May 2024 – Aug 2024*
  TikTok, San Jose, CA
  - **Data synthesis**: Generated synthetic videos to improve temporal reasoning and motion understanding in VideoLLMs.
  - **Video Encoder Optimization**: Refined the masking module to eliminate redundancy and reduce token usage for efficient motion-aware understanding.
- **Machine Learning Engineer** *Mar 2021 – Jul 2023*
  TikTok, Beijing, China
  - **Video Violation Detection with VLMs**: Developed a vision–language moderation framework to identify infrequent or long-tail policy violations in video data.
  - **Video–Text Classification & Generation**: Developed DeBERTa-based classifiers and generators for multi-video Q&A, scaling content generation to over 100M high-quality pairs.
  - **Symbolic Computation Models**: Designed transformer architectures with novel Huffman embeddings and sub-tree pretraining to improve mathematical problem-solving.

Honors & Awards

- **Silver Reviewer Award**, ICML *2026*
- **Student Conference Travel Award**, ICCV *2025*
- **Dean's Fellowship and Chair's Fellowship Award**, University of Maryland, College Park *2023*

Academic Service

- **Program Committee**, AAAI *2026*
- **Reviewer**, ICML *2026*